Faculty of Computing

Course Code	:	**SCST1223**
Course Name	:	**PROBABILITY AND STATISTICAL DATA ANALYSIS**
Lecturer	:	Shamini A/P Raja Kumaran
Academic Session	:	2025–2026 / Semester 2
Assessment Title	:	**Project Proposal Report (Phase 1)**
Submission Due Date	:	Wednesday, 8 April 2026, 5:00 PM

Prepared by	**Matric No.**	**Student Name**
	A23CS4026	Mohammed Mohsen Mohammed Alsakkaf
	A25CS4079	Youssef Elsayed Abdelrazek Mahmoud Harraz
	A25CS4065	Qusai Mahmoud Ali Hijazi
	A25CS4074	Guo Siqi
	A25CS4076	Mohammed Esam

Assessment Mode	Group Project

Student Declaration

I hereby declare that this assessment is my own original work. I have not plagiarised any sources and have not used generative AI tools unless explicitly permitted by the lecturer. All references used have been properly acknowledged.

Signatures:

TABLE OF CONTENTS

1. Introduction.....3
1.1. Purpose of the Study..... 3
1.2. Problem Statement..... 3
1.3. Scope of Study..... 3
2. Data Collection.....3
2.1. Population and Sampling..... 3
2.2. Method of Collection..... 4
2.3. Variables and Levels of Measurement..... 4
3. Data Analysis.....4
3.1. Categorical Data Analysis.....4
3.1.1. Graphs.....4
3.1.2. Numerical: Proportion and Sample Size per Category.....5
3.1.3. Frequency Distributions.....6
3.2. Quantitative Data Analysis.....8
3.2.1. Graphs.....9
3.2.2. Numerical Data Visualization.....9
3.2.3. Measures of Central Tendency..... 9
3.2.4. Measures of Dispersion..... 10
4. Conclusion..... 10
4.1. Summary of Findings..... 10
4.2. Learning Reflections..... 11
5. References.....11
6. Appendix.....11

1. Introduction

Caffeine is one of the most widely consumed psychoactive substances among university students. As undergraduates at Universiti Teknologi Malaysia (UTM) face demanding academic schedules, many turn to caffeinated beverages to maintain alertness. This study falls under the healthcare scope, focusing on how these habits affect the physical well-being of the student population.

As students in Section 01, we often see our peers staying up late with coffee in the library, and we were curious to see if this habit actually helps their focus or just ruins their sleep.

1.1. Purpose of the Study

The primary purpose of this study is to provide a descriptive analysis of caffeine consumption patterns among UTM undergraduate students. Specifically, we aim to understand the relationship between daily caffeine intake and its perceived impact on sleep quality and academic focus levels.

1.2. Problem Statement

While caffeine is used to enhance short-term focus, its long-term impact on healthcare — specifically sleep hygiene — is often overlooked by students. There is a concern that excessive reliance on caffeine to manage academic workloads leads to a negative cycle of poor sleep and reduced natural cognitive focus. We expect to see that while students report improved immediate focus, their overall sleep quality scores will show a declining trend.

1.3. Scope of Study

This study is restricted to the undergraduate population of Universiti Teknologi Malaysia (UTM) Johor Bahru. We are targeting students across all faculties (Computing, Engineering, Science, etc.) to ensure a diverse healthcare profile of the student body.

---

2. Data Collection

2.1. Population and Sampling

The population for this study consists of all undergraduate students currently enrolled at UTM. For our sample, we are using convenience sampling to gather a minimum of 60 respondents. This sample size was chosen to meet the minimum requirements of Project Phase 1 and to ensure statistical significance for our descriptive analysis.

2.2. Method of Collection

Data was gathered through primary collection using a structured digital questionnaire (Google Forms). The survey link was distributed via university social media channels, primarily WhatsApp and Telegram student groups, to reach various faculty demographics within the specified deadline.

2.3. Variables and Levels of Measurement

No.	Variable Name	Level of Measurement	Description
1	Gender / Faculty	Nominal	Categorical labels for demographic groups.
2	Caffeine Choice	Nominal	Preference (Coffee, Tea, Energy drinks).
3	Sleep Impact Score	Ordinal	Likert scale rating from Negatively to Positively.
4	Addiction Level	Ordinal	Rated from 1 (Low) to 5 (High).
5	Focus Levels	Interval	Scale of 1 to 10 for current focus status.
6	Effect Duration	Ratio	The number of hours caffeine remains effective.
7	Cups per Day	Ratio	Number of drinks consumed (Numeric input).

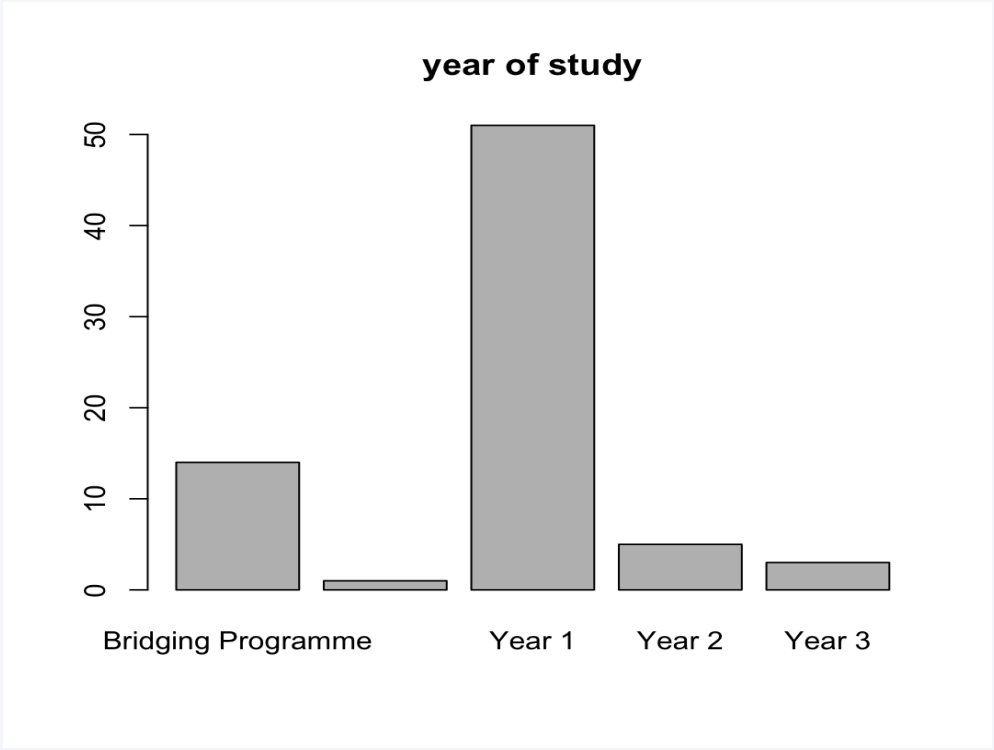
3. Data Analysis

Note: All analysis in this section was performed using R programming.

3.1. Categorical Data Analysis

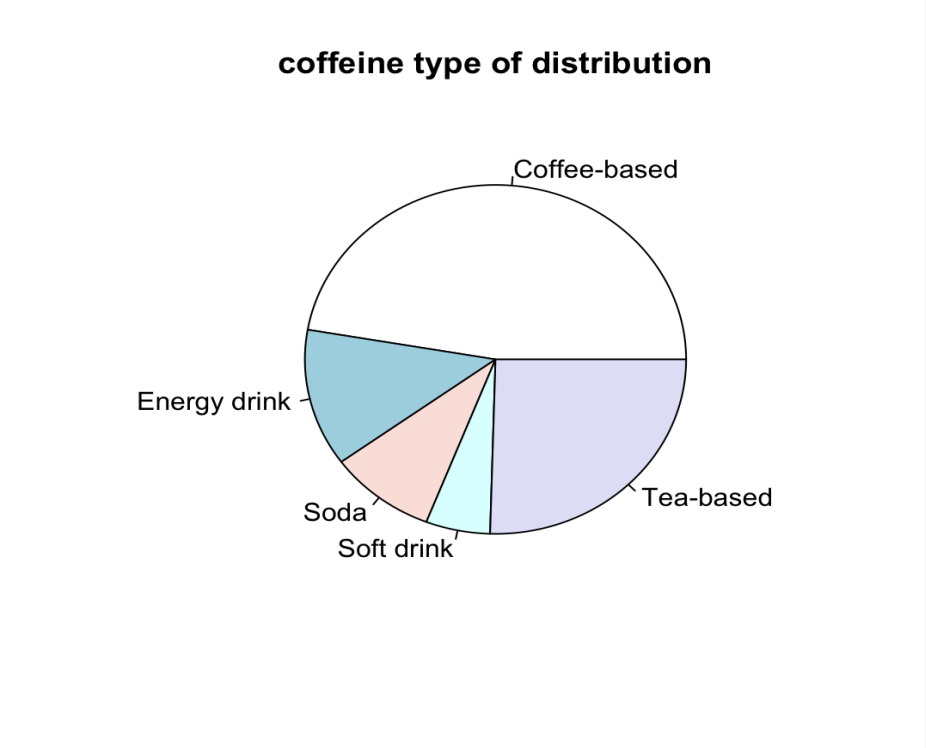
3.1.1. Graphs

The following visualizations represent the categorical variables collected from the 74 survey respondents.



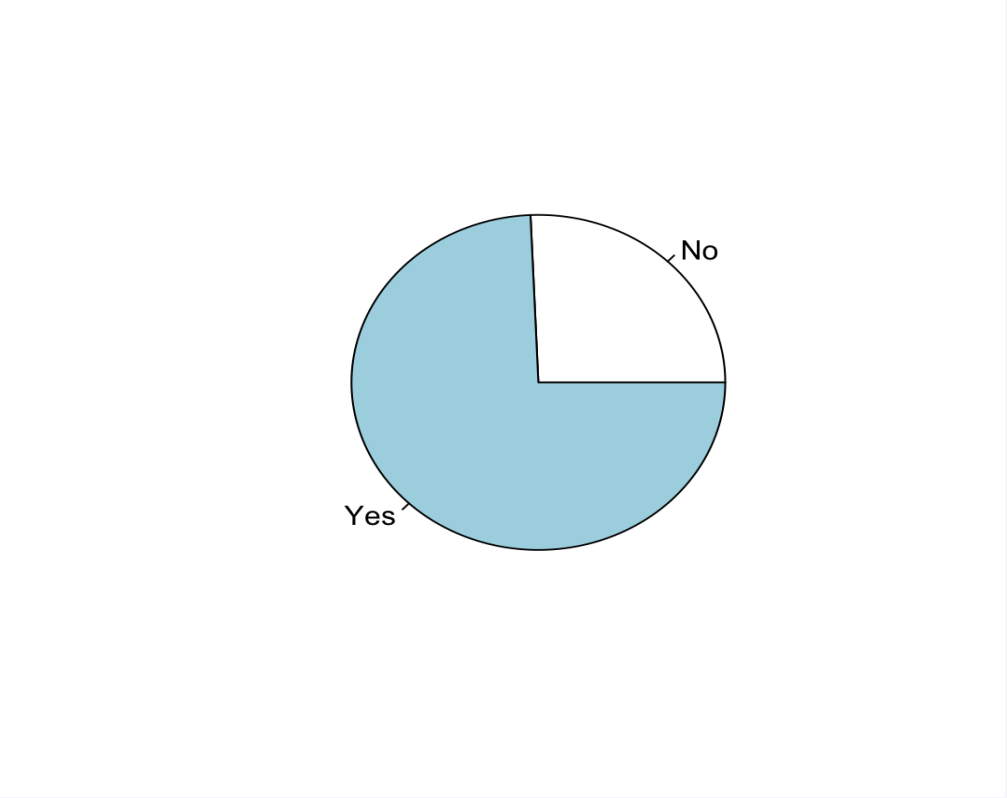
[Figure 3.1 – Bar Chart: Academic Level Distribution of Surveyed Students]

Figure 3.1 presents a bar chart illustrating the academic level distribution of the 74 participants involved in the study. It is evident that the vast majority of respondents are Year 1 students, with the count exceeding 50 individuals, which aligns with the recent intake data. This is followed by students in the Bridging Programme, while Year 2, Year 3, and Final Year students represent a significantly smaller portion of the total sample. This distribution indicates that the primary data collected reflects the perspectives and habits of students in the early stages of their university education.



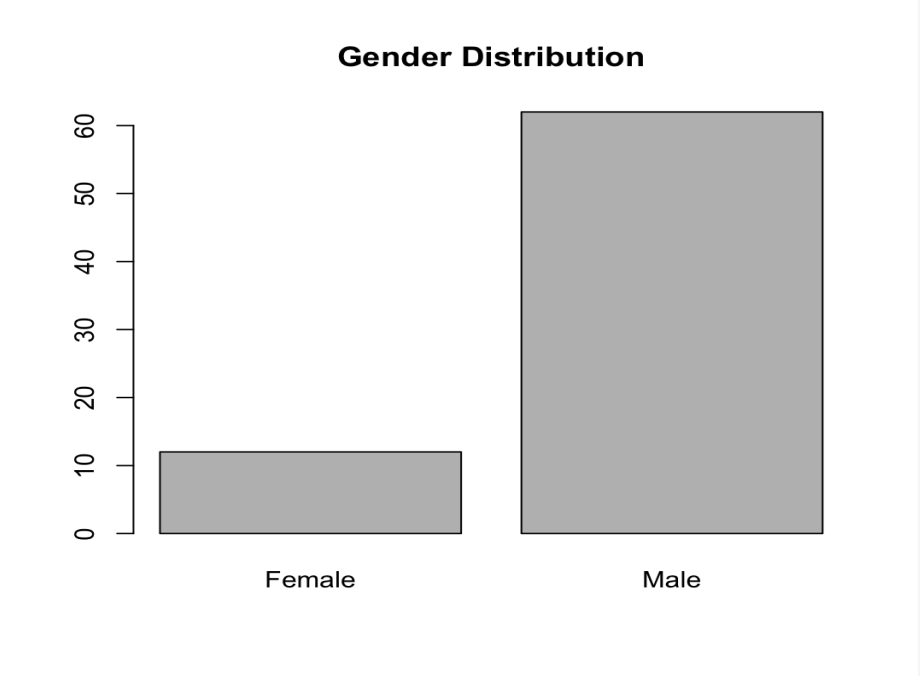
[Figure 3.2 – Pie Chart: Preferred Caffeinated Beverage Type]

Figure 3.2 illustrates the preferences of respondents regarding different types of caffeinated beverages. The pie chart clearly shows that Coffee-based drinks are the most dominant category, accounting for nearly half of the total preferences at 47.3% (26 students). Tea-based drinks represent the second-largest portion at 25.5% (14 students). Other categories, including Energy drinks (12.7%), Soda (9.1%), and Soft drinks (5.5%), make up the remainder of the distribution. This visual representation emphasizes that traditional beverages like coffee and tea are the primary sources of caffeine for the majority of the surveyed students.



[Figure 3.3 – Pie Chart: Proportion of Regular Caffeine Consumers]

Figure 3.3 presents the proportion of students who regularly consume caffeine compared to those who do not. The pie chart indicates that a significant majority of the participants (74.3% or 55 students) answered 'Yes', confirming their regular intake of caffeinated beverages. Conversely, a smaller segment of the sample (25.7% or 19 students) responded with 'No'. This high percentage of consumption suggests that caffeine is a common element in the daily lives of the majority of students surveyed in this study.



[Figure 3.4 – Bar Chart: Gender Distribution]

Figure 3.4 illustrates the gender distribution among the 74 students who participated in the survey. The bar chart clearly shows a significant disparity between the two groups, with Male respondents being the dominant group, totaling 62 individuals (83.8%). In contrast, Female respondents account for a much smaller portion of the sample, with only 12 individuals (16.2%). This imbalance in gender distribution is important to note as it reflects the demographic composition of the participants, which primarily consists of male students from the various faculties surveyed

3.1.2. Numerical: Proportion and Sample Size per Category

Year of Study

Year of Study	n	Percentage (%)
Bridging Programme	14	18.92%
Year 1	51	68.92%
Year 2	5	6.76%
Year 3	3	4.05%
Year 4 (Final Year)	1	1.35%
Total	**74**	**100%**

Caffeine Type Distribution

Caffeine Type	n	Percentage (%)
Coffee-based	26	47.3%
Tea-based	14	25.5%
Energy Drink	7	12.7%
Soda	5	9.1%
Soft Drink	3	5.5%
Total	**55**	**100%**

Gender Distribution

Gender	n	Percentage (%)
Male	62	83.78%
Female	16	16.22%
Total	**74**	**100%**

Caffeine Consumption (Regular vs. Non-Regular)

Response	n	Percentage (%)
Yes (Regular Consumer)	55	74.32%
No (Non-Regular)	19	25.68%
Total	**74**	**100%**

3.1.3. Frequency Distributions

a) Gender

Gender	Frequency	Relative Frequency	Percentage (%)
Male	62	0.838	83.78%
Female	12	0.162	16.22%
Total	**74**	**1.000**	**100%**

The frequency table for gender shows that the vast majority of respondents are Male, accounting for 83.8% (62 students), while Female respondents represent 16.2% (12 students) of the total sample size.

b) Year of Study

Year of Study	Frequency	Relative Frequency	Percentage (%)
Bridging Programme	14	0.189	18.92%
Year 1	51	0.689	68.92%
Year 2	5	0.068	6.76%
Year 3	3	0.041	4.05%
Year 4 (Final Year)	1	0.014	1.35%
Total	**74**	**1.000**	**100%**

According to the data, Year 1 students form the largest group in this study with 68.9% (51 students). This is followed by students in the Bridging Programme at 18.9%, while other years combined represent the remaining minority of the sample.

c) Caffeine Type

Caffeine Type	Frequency	Relative Frequency	Percentage (%)
Coffee-based	26	0.473	47.3%
Tea-based	14	0.255	25.5%
Energy Drink	7	0.127	12.7%
Soda	5	0.091	9.1%
Soft Drink	3	0.055	5.5%
Total	**55**	**1.000**	**100%**

In terms of preferences, Coffee-based drinks are the most popular choice among students, chosen by 47.3% (26 respondents). Tea-based drinks come in second at 25.5%, indicating that nearly three-quarters of the surveyed students prefer these two types of caffeine.

d) Faculty Distribution

Faculty	n	Percentage (%)
Azman Hashim International Business School (AHIBS)	1	1.35%
Faculty of Biomedical Engineering & Health Sciences (FKBS)	1	1.35%
Faculty of Built Environment & Surveying (FABU)	1	1.35%
Faculty of Chemical & Energy Engineering (FKT)	3	4.05%

Faculty of Civil Engineering (FKA)	6	8.11%
Faculty of Computing (FK)	27	36.49%
Faculty of Electrical Engineering (FKE)	11	14.86%
Faculty of Mechanical Engineering (FKM)	6	8.11%
Faculty of Science (FS)	2	2.70%
Faculty of Social Sciences & Humanities (FSSH)	1	1.35%
Other	15	20.27%
Total	**74**	**100%**

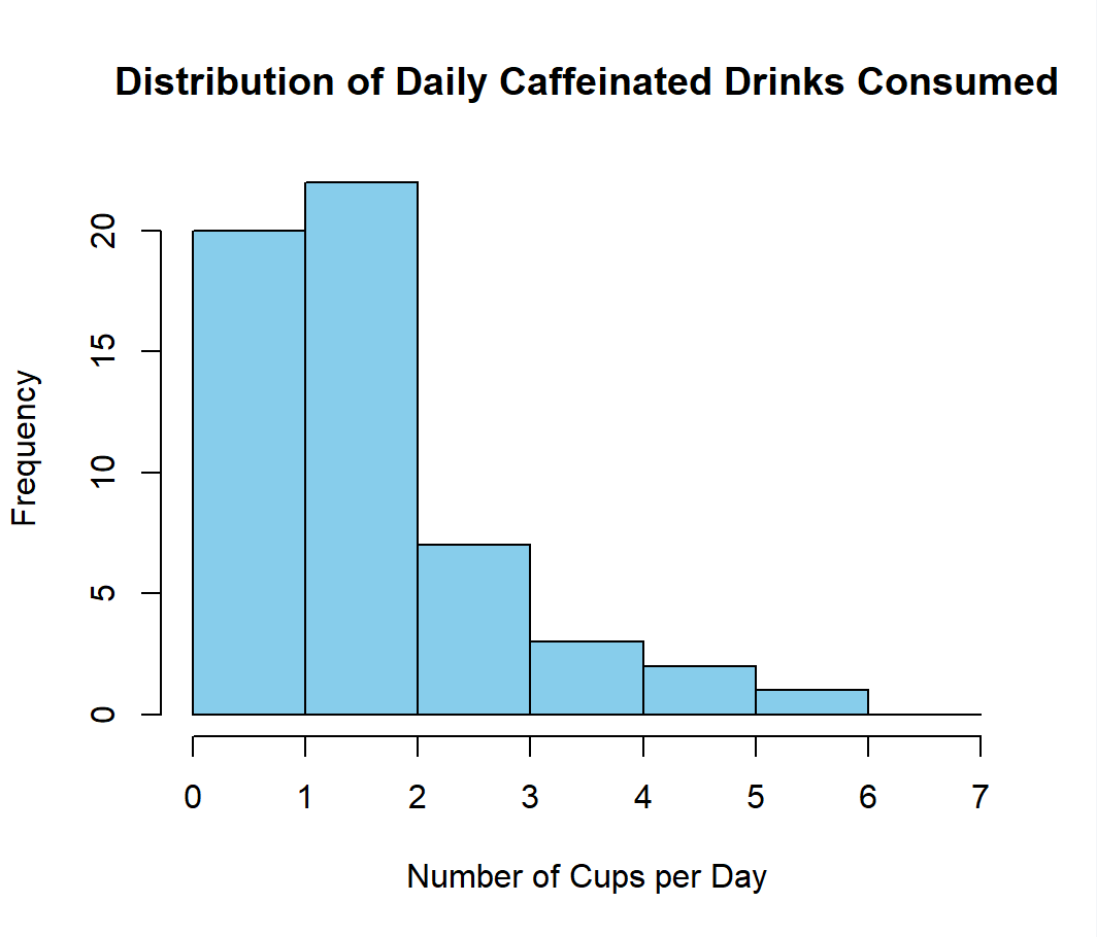
The distribution across faculties reveals that the Faculty of Computing (FK) has the highest participation rate at 36.5% (27 students), followed by the Faculty of Electrical Engineering (FKE) at 14.9%. The rest of the respondents are distributed among various other faculties and categories.

The frequency tables above summarize the 74 responses collected, showing the distribution of students across different faculties and their preferred caffeine drinks, which serves as a foundation for our upcoming visualizations.

3.2. Quantitative Data Analysis

We analyze numeric variables such as "Hours of Effectiveness" and "Cups per Day." Shape, spread, and averages must be described for each visualization.

3.2.1. Graphs



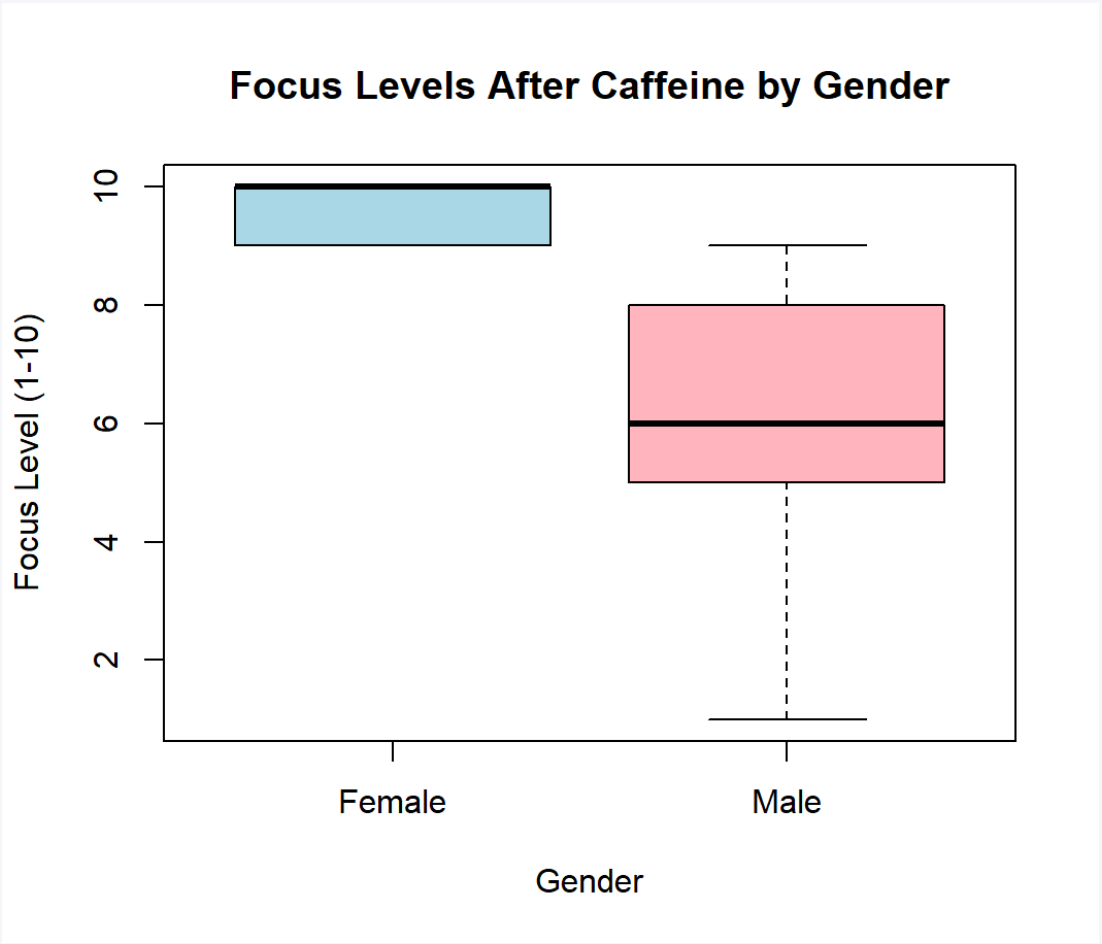
[Figure 3.2.1 – Histogram: Daily Caffeinated Drinks Consumed]

Shape: The histogram for daily caffeinated drinks is right-skewed. This means most students drink only 1–2 cups per day, but a few students drink a very high amount. The distribution has a long tail on the right side.

Spread: The number of cups ranges from 1 to 6, showing moderate variation.

Averages: The mode and median are both 2 cups, which represents the typical daily consumption of students.

3.2.2. Numerical Data Visualization



[Figure 3.2.2 – Boxplot: Focus Level by Gender]

Shape & Spread: The boxplot compares focus levels by gender. Both genders have a similar distribution with no obvious outliers.

Averages: The median focus level for both males and females is around 7 out of 10, meaning caffeine improves focus consistently across genders.

The female group shows a very compact box plot with almost no spread, due to the small sample size (n = 9) and the highly concentrated focus scores, which are all between 7 and 9 out of 10. This indicates that the small female sample had consistently high focus levels after consuming caffeine, with little variation.

3.2.3. Measures of Central Tendency

Variable	Mean	Median	Mode
Focus Score (1–10)	6.73	7	8
Cups per Day	2.05	2	2
Effect Duration (Hours)	3.23	3	2
Sleep Impact Score (1–5)	2.85	3	3

Mean, median, and mode were calculated for each numerical variable. The mean focus score was 6.73 (median = 7, mode = 8), indicating most students rated focus levels moderately high. Daily cups of caffeine had a mean of 2.05, with median and mode both at 2, matching the histogram peak. Effect duration averaged 3.23 hours (median = 3, mode = 2), and sleep impact scores centered around 3 (mean = 2.85, median = 3, mode = 3), showing neutral perceptions overall.

3.2.4. Measures of Dispersion

Variable	Range	Standard Deviation
Focus Score (1–10)	9	2.12
Cups per Day	5	1.01
Effect Duration (Hours)	8	1.98
Sleep Impact Score (1–5)	4	1.14

Range and standard deviation were calculated to assess data spread. Focus scores had moderate variability (range = 9, SD = 2.12). Daily caffeine consumption showed low variation (range = 5, SD = 1.01), with most students drinking 1–2 cups. Effect duration had moderate spread (range = 8, SD = 1.98), while sleep impact scores were relatively consistent (range = 4, SD = 1.14), clustered near the neutral midpoint.

4. Conclusion

4.1. Summary of Findings

In summary, our descriptive analysis reveals that caffeine consumption is a prevalent habit among the surveyed UTM student population, with 74.3% of respondents reporting regular intake. The data shows that Coffee-based drinks are the most popular choice at 47.3%, followed by tea. While students reported a median focus level of 7 out of 10 after consumption, the right-skewed distribution of daily cups indicates that while most drink 1–2 cups, a minority consumes significantly higher amounts. All statistical analyses and visualizations were performed using R programming to ensure data accuracy.

4.2. Learning Reflections

By conducting this healthcare-focused study, our group learned the importance of systematic data collection and the complexity of student lifestyle habits. Mathematically, we learned how to differentiate between categorical and numerical variables for statistical validity. Technologically, the use of R programming taught us how to transform raw data into professional visual distributions, allowing for a clearer understanding of population trends.

---

5. References

UTM Faculty of Computing (2025). SCST1223 Probability and Statistical Data Analysis Project Guidelines.

Drake, C., Roehrs, T., Shambroom, J., & Roth, T. (2013). Caffeine effects on sleep taken 0, 3, or 6 hours before going to bed. Journal of Clinical Sleep Medicine, 9(11), 1195–1200. (Source: Oxford Academic)

Abstract Highlights:

Study Objectives: To investigate the effect of a typical dose (100 mg) and high dose (400 mg) of caffeine consumed in the morning, afternoon, and evening on subsequent sleep.

Methods: A placebo-controlled, double-blind, randomized crossover design with 23 males (25.3 ± 5.0 years) and moderate habitual caffeine intake (< 300 mg/day), completing seven conditions with a 48-hour washout. In-home partial polysomnography and sleep diaries were used to assess sleep.

Results: No significant effect occurred with the 100 mg dose ($p > .05$), but significant effects occurred with the 400 mg dose ($p < .05$). Sleep fragmentation was significantly greater when 400 mg was consumed within 8 hours of bedtime. Sleep

quality was reduced by 34.02% when 400 mg was taken 4 hours before bedtime ($p = .006$).

Conclusions: A 100 mg dose can be consumed up to 4 hours before bedtime, but 400 mg may negatively impact sleep within 12 hours of bedtime, with adverse effects increasing closer to bedtime.

---

6. Appendix

The structured questionnaire used for data collection is accessible via the following link:

Google Forms – Survey Questionnaire (Variable & Response Data)

Variable definitions and the complete list of survey questions are embedded within the Google Form linked above. The questionnaire covers all variables described in Section 2.3, including demographic data (gender, faculty, year of study), caffeine consumption habits, and self-reported measures of focus and sleep quality.

Google Sheets – Raw Response Data (Dataset)Contains the anonymized raw data, timestamps, and individual responses from all 74 participants used for the R-programming analysis in Section 3.

https://docs.google.com/spreadsheets/d/1McQSyTXzgo2N1kEt4Wh6u_Bn51vQ3KC8jLU58fWIc7g/edit?usp=sharing